American Computer Science League

2022-2023 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$(\overline{A} + \overline{B})(B + \overline{C}) + \overline{A\,B\,C}$$

A. 7
B. 5
C. 3
D. 1
E. 0

2. Boolean Algebra

Simplify the following Boolean expression:

$$(A \oplus BC)(\overline{A}B \oplus C)$$

A. $A\overline{B}C$
B. $\overline{A}BC$
C. $AB\overline{C}$
D. $A + \overline{B} + C$
E. $AB + BC$

3. Data Structures

Build the binary search tree for the following string:

B E L M O N T S T A K E S

What is its internal path length?

A. 8
B. 46
C. 59
D. 62
E. 72

4. Data Structures

Given an initially empty queue and the following commands on the queue, what item will be popped next?

PUSH("J"), PUSH("U"), PUSH("L"), X=POP(), X=POP(), PUSH("I"), PUSH("U"), X=POP(), X=POP(), PUSH("S"), X=POP(), PUSH("C"), PUSH("A"), PUSH("E"), X=POP(), X=POP(), PUSH("S"), X=POP(), X=POP(), PUSH("A"), X=POP(), PUSH("R"), X=POP()

A. C
B. E
C. S
D. A
E. R

5. FSAs and Regular Expressions

Given the following regular expression, how many in the list do NOT satisfy it?

[a-m]*[^emtsy]*t.*y

entry	dotty	ditsy	crusty
gently	glinty	topiary	occipitally
documentary	mightily	intestacy	slapstick

A. 2
B. 3
C. 4
D. 5
E. 6